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Quantum Field Theory 1

Homework 3

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1 Exercise 3.8

The commutation relations for Dirac field in analogy with the Klein-Gordon field is

$$[\psi_a(x), \psi_b^\dagger(y)] = \delta^{(3)}(\mathbf{x}-\mathbf{y})\delta_{ab} \qquad \text{(equal times)} \tag{1}$$

where $\psi_a(x)$ is the plane wave solution to the Dirac equation and a and b denote the spinor components of ϕ. By expanding in a suitable basis the field becomes.

$$\psi(\mathbf{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\sum_{s=1,2}(a_{\mathbf{p}}^s u^s(p)e^{-ip\cdot x} + b_{\mathbf{p}}^s v^s(p)e^{ip\cdot x}) \tag{2}$$

with the adjoint field being

$$\bar{\psi}(\mathbf{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\sum_{s=1,2}(a_{\mathbf{p}}^{s\dagger} \bar{u}^s(p)e^{ip\cdot x} + b_{\mathbf{p}}^{s\dagger} \bar{v}^s(p)e^{-ip\cdot x}) \tag{3}$$

We drop the bold faced $\mathbf{p}$ from the index of energy and creation and annihilation operators for simpler notation. It is understood that this doesn't really matter since fixing the 3-momentum also fixes the 4-momentum when mass known or both fields have same mass term which is the case here. We also assume that the creation and annihilation operators are known since the point of the exercise is to compute the Hamiltonian and not the commutation relations. As such the commutation relations are assumed to be

$$[a_p^r, a_q^{s\dagger}] = [b_p^r, b_q^{s\dagger}] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})\delta_{rs} \tag{4}$$

We start with and try to quantize the Dirac Hamiltonian which is

$$H = \int d^3x\bar{\psi}(-i\boldsymbol{\gamma}\cdot\boldsymbol{\nabla} + m)\psi \tag{5}$$

we directly substitute the equations (2) and (3) for ψ and $\bar{\psi}$ and get

$$H = \int \frac{d^3x d^3p d^3q}{2(2\pi)^6}\frac{1}{\sqrt{E_p E_q}}\sum_{s,r}(a_q^{r\dagger}\bar{u}_q^r e^{iq\cdot x} + b_q^{r\dagger}\bar{v}_q^r e^{-iq\cdot x})(-i\boldsymbol{\gamma}\cdot\boldsymbol{\nabla} + m)(a_p^s u_p^s e^{-ip\cdot x} + b_p^s v_p^s e^{ip\cdot x}) \tag{6}$$

Before multiplying the terms inside the brackets we can look at some relations to see which cross terms are zero so that we don't have to compute them. For this we need the orthogonality relations

$$\bar{u}_p^r v_p^s = \bar{v}_p^r u_p^s = 0 \tag{7}$$

and

$$u^{r\dagger}(\mathbf{p})v^r(-\mathbf{p}) = 0 \tag{8}$$

Using this our integral simplifies to

$$\begin{aligned} H = \int \frac{d^3xd^3pd^3q}{2(2\pi)^6}\frac{1}{\sqrt{E_pE_q}}\sum_{s,r} -i\bigg[& a_q^{r\dagger}\bar{u}_q^r e^{iq\cdot x}(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})a_p^s u_p^s e^{-ip\cdot x} \\ & + b_q^{r\dagger}\bar{v}_q^r e^{-iq\cdot x}(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})a_p^s u_p^s e^{-ip\cdot x} + a_q^{r\dagger}\bar{u}_q^r e^{iq\cdot x}(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})b_p^s v_p^s e^{ip\cdot x} \\ & + b_q^{r\dagger}\bar{v}_q^r e^{-iq\cdot x}(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})b_p^s v_p^s e^{ip\cdot x}\bigg] + m\left(a_q^{r\dagger}a_p^s\bar{u}_q^r u_p^s e^{i(q-p)x} + b_q^{r\dagger}b_p^s\bar{v}_q^r v_p^s e^{i(p-q)x}\right) \end{aligned} \tag{9}$$

With the help of Dirac equation we get

$$-i\boldsymbol{\gamma}\cdot\boldsymbol{\nabla}u_p^s e^{-ip\cdot x} = (m - i\gamma^0\partial_0)u_p^s e^{-ip\cdot x} = (m + i^2E_p\gamma^0)u_p^s e^{-ip\cdot x} \tag{10}$$

$$-i\boldsymbol{\gamma}\cdot\boldsymbol{\nabla}v_p^s e^{ip\cdot x} = (m - i\gamma^0\partial_0)v_p^s e^{ip\cdot x} = (m - i^2E_p\gamma^0)v_p^s e^{ip\cdot x} \tag{11}$$

This gives us

$$\begin{aligned} -i\bar{u}_q^r(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})u_p^s e^{-ip\cdot x} &= m\bar{u}_q^r u_p^s e^{-ip\cdot x} + i^2E_p\bar{u}_q^r\gamma^0 u_p^s e^{-ip\cdot x} \\ &= m\bar{u}_q^r u_p^s e^{-ip\cdot x} + i^2E_p u_q^{r\dagger}u_p^s e^{-ip\cdot x} \end{aligned} \tag{12}$$

$$\begin{aligned} -i\bar{u}_q^r(\boldsymbol{\gamma}\cdot\boldsymbol{\nabla})v_p^s e^{ip\cdot x} &= m\bar{u}_q^r v_p^s e^{ip\cdot x} - i^2E_p\bar{u}_q^r\gamma^0 v_p^s e^{ip\cdot x} \\ &= m\bar{u}_q^r v_p^s e^{ip\cdot x} - i^2E_p u_q^{r\dagger}v_p^s e^{ip\cdot x} \end{aligned} \tag{13}$$

With two more equations that we get (trivially) by swapping in $\bar{v}_q^r$ to replace $\bar{u}_q^r$ in the right hand side.

The integral then beomes

$$\begin{aligned} H = \int \frac{d^3xd^3pd^3q}{2(2\pi)^6}\frac{1}{\sqrt{E_pE_q}}\sum_{s,r}\bigg[& a_q^{r\dagger}a_p^s(m\bar{u}_q^r u_p^s + i^2E_p u_q^{r\dagger}u_p^s)e^{i(q-p)x} \\ & + b_q^{r\dagger}a_p^s(m\bar{v}_q^r u_p^s + i^2E_p v_q^{r\dagger}u_p^s)e^{-i(q+p)x} + a_q^{r\dagger}b_p^s(m\bar{u}_q^r v_p^s - i^2E_p u_q^{r\dagger}v_p^s)e^{i(q+p)x} \\ & + b_q^{r\dagger}b_p^s(m\bar{v}_q^r v_p^s - i^2E_p v_q^{r\dagger}v_p^s)e^{-i(q-p)x}\bigg] + m\left(a_q^{r\dagger}a_p^s\bar{u}_q^r u_p^s e^{i(q-p)x} + b_q^{r\dagger}b_p^s\bar{v}_q^r v_p^s e^{i(p-q)x}\right) \end{aligned} \tag{14}$$

We use the following relation to calculatr integral over d^3x:

$$\int_{-\infty}^{\infty} d^3x e^{i(\mathbf{p}-\mathbf{q})x} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{15}$$

which means

$$\int d^3x e^{\mp i(q\mp p)x} = e^{\mp i(E_p \mp E_q)t}(2\pi)^3\delta^{(3)}(\mathbf{p}\mp\mathbf{q}) \tag{16}$$

After integration over d^3x our hamiltonian becomes

$$\begin{aligned}
H = \int \frac{d^3p d^3q}{2(2\pi)^3}\frac{1}{\sqrt{E_p E_q}}\sum_{s,r}\Big[& a_q^{r\dagger}a_p^s(m\bar{u}_q^r u_p^s + i^2 E_p u_q^{r\dagger}u_p^s)e^{i(E_q-E_p)t}\delta^{(3)}(\mathbf{p}-\mathbf{q}) \\
& + b_q^{r\dagger}a_p^s(m\bar{v}_q^r u_p^s + i^2 E_p v_q^{r\dagger}u_p^s)e^{-i(E_q+E_p)t}\delta^{(3)}(\mathbf{p}+\mathbf{q}) \\
& + a_q^{r\dagger}b_p^s(m\bar{u}_q^r v_p^s - i^2 E_p u_q^{r\dagger}v_p^s)e^{i(E_q+E_p)t}\delta^{(3)}(\mathbf{p}+\mathbf{q}) \\
& + b_q^{r\dagger}b_p^s(m\bar{v}_q^r v_p^s - i^2 E_p v_q^{r\dagger}v_p^s)e^{-i(E_q-E_p)t}\delta^{(3)}(\mathbf{p}-\mathbf{q})\Big] \\
& + m\left(a_q^{r\dagger}a_p^s\bar{u}_q^r u_p^s e^{i(E_q-E_p)t}\delta^{(3)}(\mathbf{p}-\mathbf{q}) + b_q^{r\dagger}b_p^s\bar{v}_q^r v_p^s e^{-i(E_q-E_p)t}\delta^{(3)}(\mathbf{p}-\mathbf{q})\right)
\end{aligned} \tag{17}$$

Now we integrate over d^3q and eliminate terms proportional to (7) and (8). Terms like $u_q^{r\dagger}v_p^s$ are eliminated by relation (8) since these terms are proportional to $\delta^{(3)}(\mathbf{p}+\mathbf{q})$ which gives $\mathbf{q}=-\mathbf{p}$ which in turn makes these terms into $u^{r\dagger}(\mathbf{p})v^s(-\mathbf{p})$. The remaining Hamiltonian is now

$$\begin{aligned}
H = \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_p}\sum_{s,r}\Big[& a_p^{r\dagger}a_p^s(m\bar{u}_p^r u_p^s + i^2 E_p u_p^{r\dagger}u_p^s) \\
& + b_p^{r\dagger}b_p^s(m\bar{v}_p^r v_p^s - i^2 E_p v_p^{r\dagger}v_p^s)\Big] + m(a_p^{r\dagger}a_p^s\bar{u}_p^r u_p^s + b_p^{r\dagger}b_p^s\bar{v}_p^r v_p^s)
\end{aligned} \tag{18}$$

$$= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_p}\sum_{s,r}\left[2m(a_p^{r\dagger}a_p^s\bar{u}_p^r u_p^s + b_p^{r\dagger}b_p^s\bar{v}_p^r v_p^s) - E_p a_p^{r\dagger}a_p^s u_p^{r\dagger}u_p^s + E_p b_p^{r\dagger}b_p^s v_p^{r\dagger}v_p^s\right] \tag{19}$$

From the normalisation of the spinors we have:

$$u_p^{r\dagger}u_p^s = 2E_p\delta^{rs} \qquad v_p^{r\dagger}v_p^s = 2E_p\delta^{rs} \tag{20}$$

$$\bar{u}_p^r u_p^s = 2m\delta^{rs} \qquad \bar{v}_p^r v_p^s = -2m\delta^{rs} \tag{21}$$

Using these we get

$$H = \int \frac{d^3p}{(2\pi)^3} \frac{1}{E_p} \sum_s \left[(2m^2 - E_p^2)(a_p^{s\dagger} a_p^s - b_p^{s\dagger} b_p^s) \right] \tag{22}$$

Here $m^2 = E_p^2$ since when normalizing the spinors Peskin & Schroeder used lorentz boost to a frame where $m^2 = E_p^2$. Putting $m^2 = E_p^2$ gives us the Hamiltonian

$$H = \int \frac{d^3p}{(2\pi)^3} \sum_s \left[E_{\mathbf{p}} a_{\mathbf{p}}^{s\dagger} a_{\mathbf{p}}^s - E_{\mathbf{p}} b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^s \right] \tag{23}$$

This looks like the hamiltonian for the free Klein-Gordon (complex) field but with negative sign on one of the creation operator. This means that with the negative creation operator a particle of negative energy can be created infinitely many times which means that the Hamiltonian is unbounded from below.

2 Exercise 3.9

Derive the Gordon identity

$$\bar{u}(p')\gamma^\mu u(p) = \bar{u}(p') \left[\frac{p'^\mu + p^\mu}{2m} + \frac{i\sigma^{\mu\nu} q_\nu}{2m} \right] u(p) \tag{24}$$

where $q = p' - p$ and $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$

$$\begin{aligned} \sigma^{\mu\nu} &= \frac{i}{2}[\gamma^\mu, \gamma^\nu] && (25)\\ &= \frac{i}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) && (26)\\ &= \frac{i}{2}(2\gamma^\mu\gamma^\nu - \{\gamma^\mu, \gamma^\nu\}) && (27)\\ \Rightarrow i\sigma^{\mu\nu} &= \eta^{\mu\nu} - \gamma^\mu\gamma^\nu && (28) \end{aligned}$$

from (26) we can also make the following substitution:

$$\sigma^{\mu\nu} = \frac{i}{2}(-2\gamma^\nu\gamma^\mu + \gamma^\mu, \gamma^\nu) \tag{29}$$

$$\Rightarrow i\sigma^{\mu\nu} = \gamma^\nu\gamma^\mu - \eta^{\mu\nu} \tag{30}$$

So we have two different definition for $\sigma^{\mu\nu}$ and they have opposite sign metric tensor which will be usefull for calculating the identity. We start the derivation with

$$\bar{u}(p')i\sigma^{\mu\nu}q_\nu u(p) = \bar{u}(p')\left[\sigma^{\mu\nu}p'_\nu - \sigma^{\mu\nu}p_\nu\right]u(p) \tag{31}$$

we use different definitons of $\sigma^{\mu\nu}$ to multiply p'_ν and p_ν

$$\bar{u}(p')i\sigma^{\mu\nu}q_\nu u(p) = \bar{u}(p')\left[(\gamma^\nu\gamma^\mu - \eta^{\mu\nu})p'_\nu - (\eta^{\mu\nu} - \gamma^\mu\gamma^\nu)p_\nu\right]u(p) \tag{32}$$

$$= \bar{u}(p')\left[(\gamma^\nu\gamma^\mu p'_\nu + \gamma^\mu\gamma^\nu p_\nu - (p'^\mu + p^\mu)\right]u(p) \tag{33}$$

$$\bar{u}(p')[(p'^\mu + p^\mu) + i\sigma^{\mu\nu}q_\nu]u(p) = \bar{u}(p')\left[(\gamma^\nu p'_\nu\gamma^\mu + \gamma^\mu\gamma^\nu p_\nu\right]u(p) \tag{34}$$

The Dirac equation and it's conjugate form are:

$$(\gamma^\mu p_\mu - m)u(p) = 0 \tag{35}$$

$$\bar{u}(p)(\gamma^\mu p_\mu - m) = 0 \tag{36}$$

Applying this on (34) we get

$$\bar{u}(p')[(p'^\mu + p^\mu) + i\sigma^{\mu\nu}q_\nu]u(p) = \bar{u}(p')\left[(m\gamma^\mu + \gamma^\mu m\right]u(p) \tag{37}$$

$$= 2m\bar{u}(p')\gamma^\mu u(p) \tag{38}$$

$$\Rightarrow \bar{u}(p')\gamma^\mu u(p) = \bar{u}(p')\left[\frac{(p'^\mu + p^\mu)}{2m} + \frac{i\sigma^{\mu\nu}q_\nu}{2m}\right]u(p) \quad \square \tag{39}$$

3 Exercise 3.10

Investigate the transformation properties of $\bar{\psi}(x)\gamma^\mu\gamma_5\psi(x)$

This spinor field and it's adjoint field transform as

$$\psi \to \Lambda_{\frac{1}{2}}\psi \tag{40}$$

$$\bar{\psi} \to \bar{\psi}\Lambda_{\frac{1}{2}}^{-1} \tag{41}$$

Thus we get the transformation

$$\bar{\psi}(x)\gamma^\mu\gamma_5\psi(x) \to \bar{\psi}(x)\Lambda_{\frac{1}{2}}^{-1}\gamma^\mu\gamma_5\Lambda_{\frac{1}{2}}\psi(x) \tag{42}$$

and

$$\bar{\psi}(x)\Lambda_{\frac{1}{2}}^{-1}\gamma^{\mu}\gamma_5\Lambda_{\frac{1}{2}}\psi(x) = \bar{\psi}(x)\Lambda_{\frac{1}{2}}^{-1}\gamma^{\mu}\Lambda_{\frac{1}{2}}\Lambda_{\frac{1}{2}}^{-1}\gamma_5\Lambda_{\frac{1}{2}}\psi(x) \tag{43}$$

We have

$$\Lambda_{\frac{1}{2}}^{-1}\gamma^{\mu}\Lambda_{\frac{1}{2}} = \Lambda^{\mu}_{\nu}\gamma^{\nu} \tag{44}$$

from exercise 3.3 so now we only need to know the transformation $\Lambda_{\frac{1}{2}}^{-1}\gamma_5\Lambda_{\frac{1}{2}}$

$$\Lambda_{\frac{1}{2}}^{-1}\gamma_5\Lambda_{\frac{1}{2}} = \Lambda_{\frac{1}{2}}^{-1}\frac{-i}{4!}\epsilon_{\mu\nu\rho\sigma}\Lambda_{\frac{1}{2}}\gamma^{\mu}\gamma^{\nu}\gamma^{\rho}\gamma^{\sigma}\Lambda_{\frac{1}{2}} \tag{45}$$

$$= -\frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\left[\Lambda_{\frac{1}{2}}^{-1}\gamma^{\mu}\Lambda_{\frac{1}{2}}\Lambda_{\frac{1}{2}}^{-1}\gamma^{\nu}\Lambda_{\frac{1}{2}}\Lambda_{\frac{1}{2}}^{-1}\gamma^{\rho}\Lambda_{\frac{1}{2}}\Lambda_{\frac{1}{2}}^{-1}\gamma^{\sigma}\Lambda_{\frac{1}{2}}\right] \tag{46}$$

we now use relation (44)

$$\Lambda_{\frac{1}{2}}^{-1}\gamma_5\Lambda_{\frac{1}{2}} = -\frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\left[\Lambda^{\mu}_{\lambda}\gamma^{\lambda}\Lambda^{\nu}_{\gamma}\gamma^{\gamma}\Lambda^{\rho}_{\delta}\gamma^{\delta}\Lambda^{\sigma}_{\eta}\gamma^{\eta}\right] \tag{47}$$

$$= -\frac{i}{4!}\epsilon_{\mu\nu\rho\sigma}\left[\Lambda^{\mu}_{\lambda}\Lambda^{\nu}_{\gamma}\Lambda^{\rho}_{\delta}\Lambda^{\sigma}_{\eta}\right]\gamma^{\lambda}\gamma^{\gamma}\gamma^{\delta}\gamma^{\eta} \tag{48}$$

Using the results from exersice 1.3 we get

$$\epsilon_{\mu\nu\rho\sigma}\left[\Lambda^{\mu}_{\lambda}\Lambda^{\nu}_{\gamma}\Lambda^{\rho}_{\delta}\Lambda^{\sigma}_{\eta}\right] = \epsilon_{\lambda\gamma\delta\eta}\det(\Lambda) \tag{49}$$

putting this into our transformation gives us

$$\Lambda_{\frac{1}{2}}^{-1}\gamma_5\Lambda_{\frac{1}{2}} = -\frac{i}{4!}\det(\Lambda)\epsilon_{\lambda\gamma\delta\eta}\gamma^{\lambda}\gamma^{\gamma}\gamma^{\delta}\gamma^{\eta} \tag{50}$$

$$= \det(\Lambda)\gamma_5 \tag{51}$$

Now we go back to equation (43) and substitute the results which gives us

$$\bar{\psi}(x)\Lambda_{\frac{1}{2}}^{-1}\gamma^{\mu}\gamma_5\Lambda_{\frac{1}{2}}\psi(x) = \bar{\psi}(x)\Lambda^{\mu}_{\nu}\gamma^{\nu}\det(\Lambda)\gamma_5\psi(x) \tag{52}$$

So we get the transformation

$$\bar{\psi}(x)\gamma^{\mu}\gamma_5\psi(x) \to \det(\Lambda)\bar{\psi}(x)\Lambda^{\mu}_{\nu}\gamma^{\nu}\gamma_5\psi(x) \tag{53}$$